IS1212 : Probability and Statistics

K S Saubhagya
Lecturer (Probationary)
Department of Statistics, Faculty of Science, University of Colombo
saubhagya@stat.cmb.ac.lk

Introduction to Probability

- Probabilities are associated with experiments where the outcome is not known in advance or cannot be predicted.
- An experiment is any process that can be repeated in which the results are uncertain.
- Probability measures and quantifies "how likely" an event, related to the experiment, will happen.

Terminology

- **Random experiment:** A process or activity which produces several possible outcomes. The outcomes which result cannot be predicted with absolute certainty.

E.g. Tossing a coin, rolling dice

- **Outcome**: The result(s) of a random experiment.

Sample Space

- The **sample space** is the set of all possible outcomes in an experiment.

Example: If a die is rolled, the sample space S is given by

$$S = \{1,2,3,4,5,6\}$$

Example: If two coins are tossed, the sample space S is given by

$$S = \{HH,HT,TH,TT\}$$, where H = head and T = tail

Examples

Random Experiment	Sample Space
Toss a coin	S={Heads,Tails}
Roll a die	S={1,2,3,4,5,6}
Number of goals in a soccer match	S={0,1,2,3,⋯}

- When we repeat a random experiment several times, we call each one of them a **trial.**
- In example 1 (toss a coin), each trial will result in either heads or tails.

- Sample space is defined based on how you define the random experiment.

E.g.: **Random Experiment**: We toss a coin three times and observe the sequence of heads/tails.

Sample Space: S={(H,H,H),(H,H,T),(H,T,H),(T,H,H),(H,T,T),(T,H,T),(T,T,H),(T,T,T)}

Event

- We define an **event** as **some specific outcome of an experiment**. An event is a subset of the sample space.

Example:

Suppose we randomly select a student and ask him, "how many pairs of jeans do you own?".

$$S = \{0, 1, 2, 3, ...\}$$

Define the following events.

1. Event that a randomly selected student owns no jeans.
2. Event that a randomly selected student owns some jeans.
3. Event that a randomly selected student owns no more than five pairs of jeans.
4. Event that a randomly selected student owns an odd number of pairs of jeans.

Answer

If A is the event that a randomly selected students owns no jeans:

A = student owns none = {0}

If B is the event that a randomly selected student owns some jeans:

B = student owns some = {1,2,3, ….}

If C is the event that a randomly selected student owns no more than five pairs of jeans:

C = student owns no more than five pairs = {0,1,2,3,4,5}

And, if D is the event that a randomly selected student owns an odd number of pairs of jeans:

D = student owns an odd number = {1,3,5, …. }

Independent Events

- **Independent Events:** Two events may be independent when the actual happening of one does not influence in any way the probability of the happening of the other one.

Example: Event of getting a head on the first coin and the event of getting a tail on the second coin in a simultaneous throw.

- Two events E and F are **independent** if and only if

$$\mathbf{P(E \text{ and } F) = P(E)P(F)} \quad \text{or} \quad \mathbf{P(E \cap F) = P(E)P(F)}$$

- This is the multiplication rule on the independent events.

Mutually Exclusive Events

- **Mutually Exclusive Events:** Two events are known as mutually exclusive when the occurrence of one of them excludes the occurrence of the other.

Example: Getting head or tail when a coin is tossed, but not both.

- Two events E and F are **mutually exclusive** if and only if

$$P(E \cap F) = 0$$

Addition Rule

P(AUB) = P(A) + P(B) - P(A∩B)

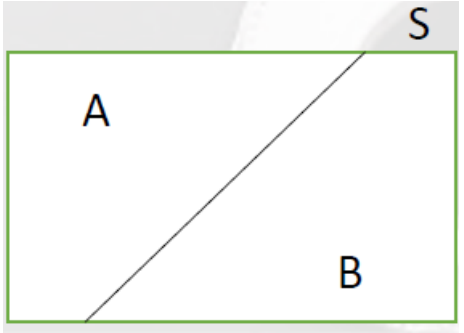
- If two events A and B are mutually exclusive (disjoint) then,

$$P\ (A \cup B) = P\ (A\ ) + P\ (B)$$

- If two events A and B are exhaustive then,

$$P(\ A \cup B) = 1$$

From above,	$P(A) + P(B) = 1$ then $P(B) = 1 - P(A)$
We know that,	$P(A') = 1 - P(A)$
Therefore,	$P(B) = P(A')$
Thus,	$B = A'$ and $A = B'$



Complement of an Event

- Let S denote the sample space of a probability experiment and let E denote an event. The **complement of E** is denoted by $\bar{E}$ **or** E' is all simple events in the sample space S that are not in the simple events E.
- If E represents any event, $\bar{E}$ represents the complement of E, then

$$P(\bar{E}) = 1 - P(E)$$

Example on Complement Rule

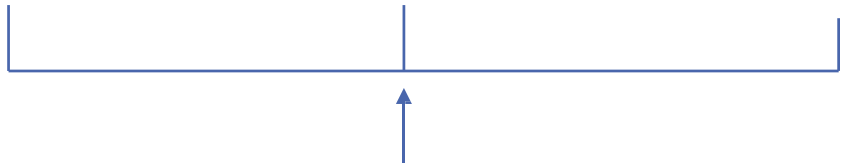
Q) According to the American Veterinary Medical Association, 31.6% of American households own a dog. What is the probability that a randomly selected household does not own a dog?

$$E = \text{Owning a dog}$$
$$P(E) = 0.316$$
$$P(E') = 1 - P(E)$$
$$= 1 - 0.316$$
$$= 0.684$$

Probability Definition (Informal)

- The value of a probability is a number between 0 and 1 inclusive.

Probability: 0 0.5 1.0



The occurrence of the event is just as likely as it is unlikely

a number close to **0** means "**not likely**"

a number close to **1** means "**quite likely**"

If the probability of an event is exactly 0, then the event can't occur.

If the probability of an event is exactly 1, then the event will definitely occur.

How to assign Probability to Events

- There are three approaches
    1. Personal Opinion approach
    2. Relative Frequency approach
    3. Classical approach

Personal Opinion Approach

"I think there is an 80% chance of rain today."

"I think there is a 50% chance that the world's oil reserves will be depleted by the year 2100."

"I think there is a 1% chance that the men's basketball team will end up in the Final Four some time this decade."

Question

At which end of the probability scale would you put the probability that:

1. one day you will die?
2. you can swim around the world in 30 hours?
3. you will win the lottery some day?
4. a randomly selected student will get an A in this course?
5. you will get an A in this course?

Relative Frequency Approach

1. Perform an experiment a large number of times, **n**, say.
2. Count the number of times the event A of interest occurs, call the number **N(A)**, say.
3. Then, the probability of event A equals:

$$P(A) = \frac{N(A)}{n}$$

Classical approach/Formal Definition of Probability

Definition: Probability is a (real-valued) set function P that assigns to each event A in the sample space S a number P(A), called the probability of the event A, such that the following hold:

- The probability of any event A must be nonnegative, that is, $P(A) \geq 0$.
- The probability of the sample space is 1, that is, $P(S) = 1$.
- Given mutually exclusive events $A_1, A_2, A_3, \ldots$ that is, where

$$A_i \cap A_j = \emptyset, \text{ for } i \neq j \text{ and}$$

$$P(A_1 \cup A_2 \cup \ldots.. \cup A_k) = P(A_1) + P(A_2) + \ldots. + P(A_k)$$

Example: The responses for a close ended question to rate the customer satisfaction is as follows.

Status	Extremely Satisfied (ES)	Satisfied (S)	Moderate (M)	Dissatisfied (D)	Extremely Dissatisfied (DS)
Count	30	50	100	20	10
Probability	0.14	0.24	0.48	0.1	0.04

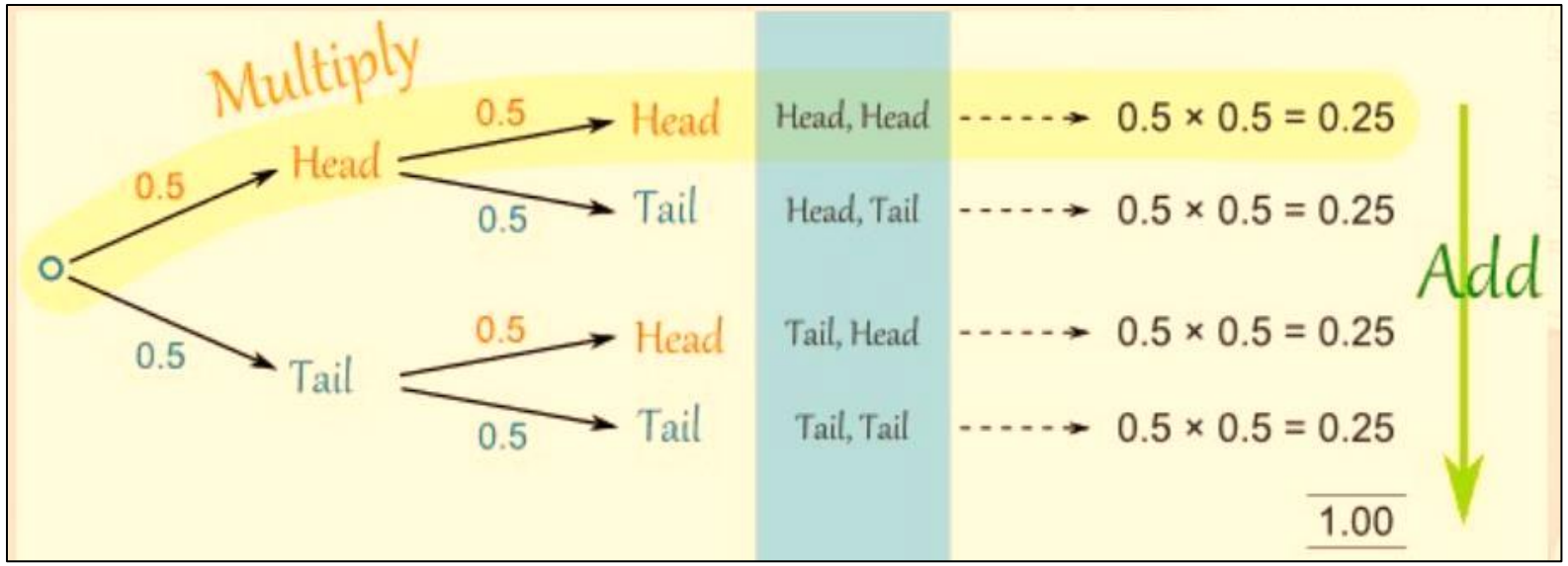
Note that the above three axioms are correct.

Theorems of Probability

- **Theorem #1:** $P(A) = 1 - P(A')$
- **Theorem #2:** $P(\varnothing) = 0$
- **Theorem #3:** If events A and B are such that $A \subseteq B$, then $P(A) \leq P(B)$
- **Theorem #4**: $0 \leq P(A) \leq 1$
- **Theorem #5:** For any two events A and B, $P(A \cup B) = P(A) + P(B) - P(A \cap B)$.
- **Theorem #6:** $P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(A \cap C) - P(B \cap C) + P(A \cap B \cap C)$

Tree diagram

- Draw a tree diagram to show probabilities associated with tossing a fair coin twice.
- Note: each toss of a coin is independent of the previous toss.



Contingency Table

- Simple Probability refers to the probability of a simple event.
  - E.g.: P(Jan.)
  - E.g.: P(Wed.)

	Jan.	Not Jan.	Total
Wed.	4	48	52
Not Wed.	27	286	313
Total	31	334	365

P(Wed.) = 52 / 365

P(Jan.) = 31 / 365

Contingency Table

- Joint Probability refers to the probability of an occurrence of two or more events (joint event).
  - E.g.: P(Jan. and Wed.)
  - E.g.: P(Not Jan. and Not Wed.)

	Jan.	Not Jan.	Total
Wed.	4	48	52
Not Wed.	27	286	313
Total	31	334	365

P(Not Jan. and Not Wed.)
= 286 / 365

P(Jan. and Wed.) = 4 / 365

Question 1

A box contains 7 red balls, 4 white balls and 9 blue balls. Find the following

probabilities

1. Getting a red ball
2. Getting a white ball
3. Getting a blue ball.
4. Getting two red balls one at a time without replacement.

Question 2

- A batch of 50 parts contains six defects. If two parts are drawn at random, one at a time without replacement, what is the probability that both parts are defective?
- If this experiment is repeated, with replacement, what is the probability that both parts are defective?

Question 3

A box contains 5 green pencils & 7 yellow pencils. Two pencils are chosen at random from the box without replacement.

1. What is the probability that they are both yellow?
2. What is the probability that they are same in color?

Question 4

In a group of 150 students 45 are first years and 55 are second years. Find the probability that a student picked from this group at random is either a first year or second year.

Question 5

In a group of 101 students 40 are juniors, 50 are female, and 22 are female juniors. Find the probability that a student picked from this group at random is either a junior or female.

Question 6

A company has bid on two large construction projects. The company president believes that the probability of winning the first contract is 0.6, the probability of winning the second contract is 0.4, and the probability of winning both contracts is 0.2.

1. What is the probability that the company wins at least one contract?

2. What is the probability that the company wins the first contract but not the second contract?

3. What is the probability that the company wins neither contract?

4. What is the probability that the company wins exactly one contract?

Any questions?